5.4 Knowledge graphs

The primary tool for implementing an ontology and building a robust, interconnected model of reality for a digital twin is the *knowledge graph* which is a structured representation of information that models real-world entities like (nodes) and the relationships (edges) between them. These relationships carry semantic meaning, making the data machine-readable and enabling automated reasoning about complex connections.

5.4.1 Graph theory

Knowledge graphs are based on graph theory, which models systems as connected points (nodes) and lines (edges). This abstraction reveals that complex problems are often about the underlying relational structure rather than granular physical details.

For example, figure 5.6 shows how the seven bridges of the city of Königsberg (now Kaliningrad) can be represented as an abstract network. The physical layout on the left becomes the graph structure on the right—nodes connected by edges. In a knowledge graph for a digital twin, these connections carry additional semantics, for example, a 'hasLocation' edge between a sensor node and a room node doesn't just show they're connected, but explicitly defines their spatial relationship.

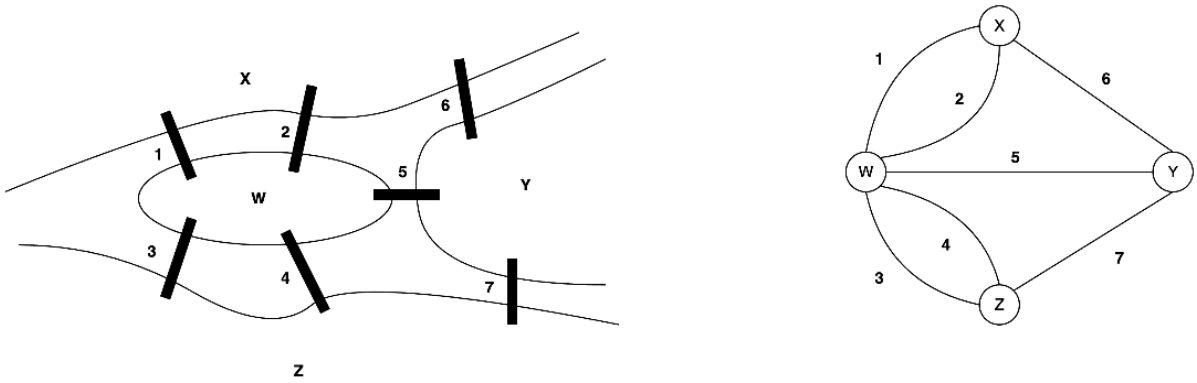


Figure 5.6 Euler's '7 Bridges of Königsberg' problem that was the origin of graph theory. He proved it was not possible to cross each of the 7 bridges connecting the two islands in the city (shown to the left) once and only once. The abstract representation of the seven bridges, two islands, and the shore either side of the river as nodes connected by edges is shown to the right.